

Product Laws of Exponents*Question Bank 3.01*

Show your work. Circle your final answer.

For questions 1 and 2, write the product as a repeated multiplication expression, then as a single power.

1. $3^3 \cdot 3^4$

2. $5^1 \cdot 5^4 \cdot 5^3$

3. Evaluate each power, then multiply. $4^1 \cdot 4^3$

4. Apply the product of powers law, then evaluate the power. $(-8)^2 \cdot (-8)^6$

5. Jose says he can evaluate $9^3 \cdot 9^{12}$ using the product of powers law. Is he correct? Explain, and write the single power.

For questions 6 through 8, use the power of a product law.

6. Write $(4 \cdot 5)^2$ as a repeated multiplication expression, group the repeated factors, then write it in the form $a^m \cdot b^n$.

7. Evaluate inside the parentheses first, then apply the exponent. $(-7 \cdot 5)^4$

8. Apply the power of a product law, then evaluate. $(3 \cdot (-6))^2$

For questions 9 through 11, use the power of a power law or the identity law.

9. Write $[(3)^2]^8$ as a single power.

10. Rewrite $(0.5 \cdot (-8))^2$ using a law of exponents, name the law, then evaluate.

11. Write each expression using the identity law, then give its value.

a. 285.6^1

b. $\left(\frac{24}{8}\right)^1$

Working backwards.

12. Find the value of n that makes each statement true.

a. $2^3 \cdot 2^n = 2^8$

b. $(5^2)^n = 5^{10}$

c. $(7 \cdot 3)^n = 7^4 \cdot 3^4$

13. Which expression is equivalent to $6^2 \cdot 6^5$?

- ☐ A. 6^7
- ☐ B. 12^7
- ☐ C. 36^7
- ☐ D. 6^{10}

14. Select ALL of the expressions that are equivalent to 2^6 .

- ☐ A. 4^3
- ☐ B. $2^3 \cdot 2^3$
- ☐ C. $(2^3)^2$
- ☐ D. $(2^2)^4$
- ☐ E. $2^2 \cdot 2^4$
- ☐ F. $2^2 \cdot 2^3$